

CS & IT ENGINEERING



Computer Network

Subnetting

Lecture No. - 05



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Recap of Previous Lecture



Topic

CIDR Notation





Topics to be Covered



Topic

Subnetting



[MCQ] [GATE-2010] [2-Mark] IIT-G



#Q. Suppose computers A and B have IP addresses "10.105.1.113" and "10.105.1.91" respectively and they both use the same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network?

~~A~~ 255.255.255.0 → 24 bits

~~B~~ 255.255.255.128 → 25 bits

~~C~~ 255.255.255.192 → 26 bits

☒ D 255.255.255.224 → 27 bits

Ans: D

⑦

10.105.1.113 $\longrightarrow$ = 10.105.1.01110001

10.105.1.91 $\longrightarrow$ = 10.105.1.01011011

255.255.255.224

= 255.255.255.11100000

27 bit NetID

5 bit HostID

c

10.105.1.113

=

10.105.1.01110001

10.105.1.91

=

10.105.1.01011011

255.255.255.192

=

255.255.255.11000000

26 bit NetID

10 . 105 . 1 . 113 = 10 . 105 . 1 . 0 1 1 1 0 0 0 1

10 . 105 . 1 . 91 = 10 . 105 . 1 . 0 1 0 1 1 0 1 1

255 . 255 . 255 . 128 = 255 . 255 . 255 . 1 0 0 0 0 0 0 0

10 . 105 . 1 . 113 = 10 . 105 . 1 . 113

10 . 105 . 1 . 91 = 10 . 105 . 1 . 91

255 . 255 . 255 . 0 = 255 . 255 . 255 . 0



Topic : Subnetting



→ Dividing (**logically**) a **network** into smaller manageable **sub-networks**



Topic : Subnetting



→ **Sub-network (subnet)** : **Clustering of hosts inside a network**





Topic : Subnetting

$\Rightarrow$ CIDR



→ Clustering of hosts based on most significant bits of host identifier





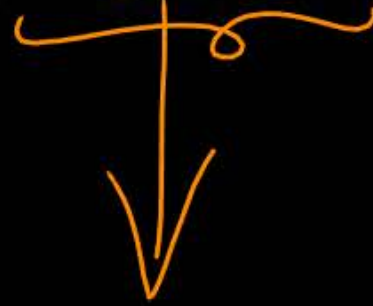
Topic : Subnetting



→ Before subnetting, IP address having two sections :

1. Network Identifier (Net ID) : x bits

2. Host Identifier (Host ID) : y bits





Topic : Subnetting



→ After subnetting, IP address having three sections :

1. Network Identifier (Net ID) : x bits
2. Sub-network Identifier (Subnet ID) : y₁ bits
3. Host Identifier (Host ID) : y₂ bits

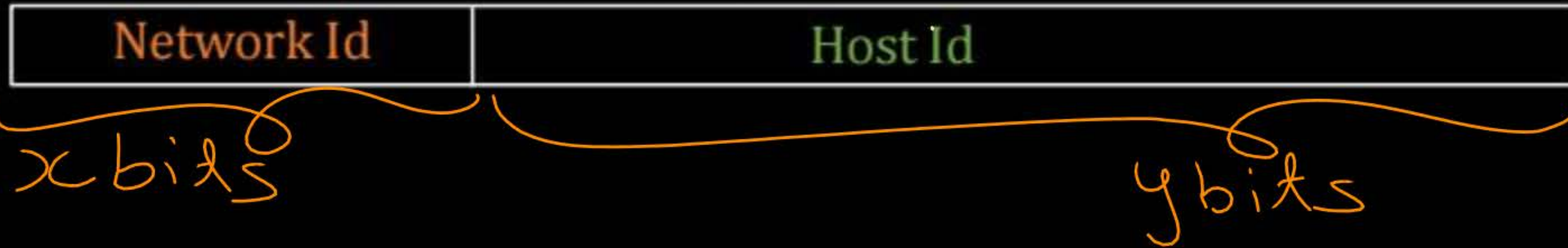


Topic : Subnetting

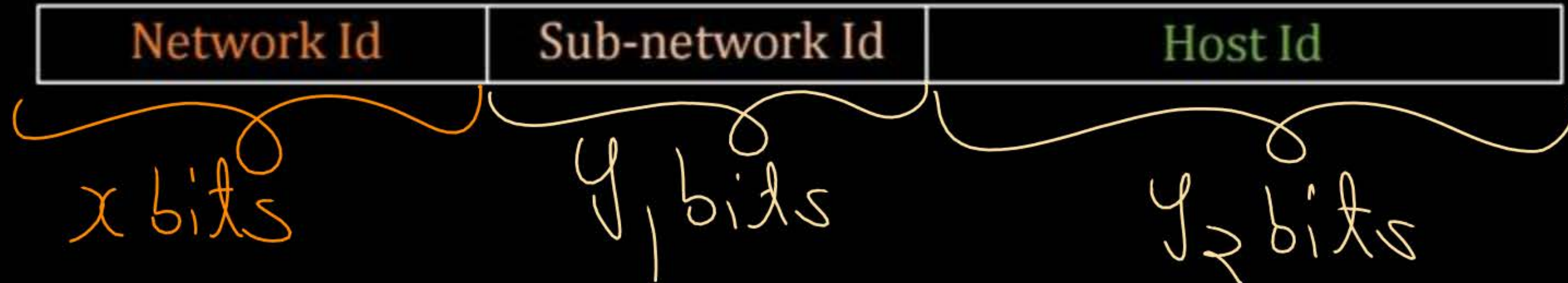
$$(y = y_1 + y_2)$$



Before Subnetting



After Subnetting





Topic : Subnet Address



→ Used to represent a sub-network

→ Special IP address (32 bits)

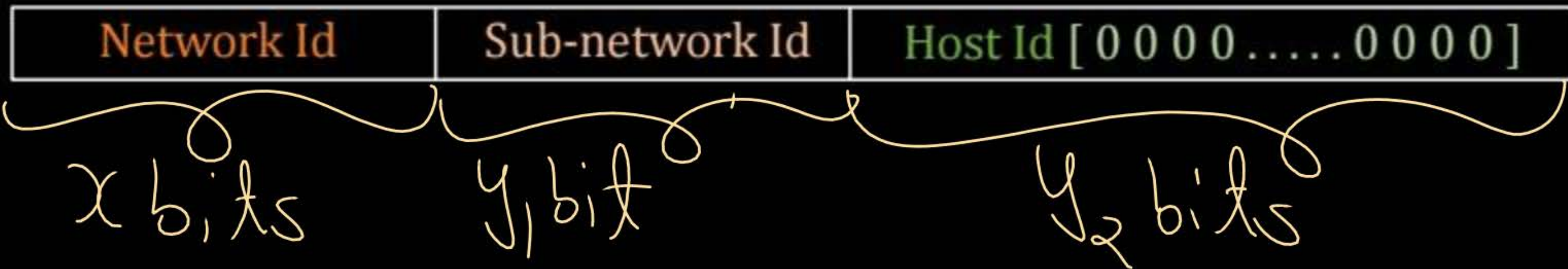


Topic : Subnet Address

NetID field = Assigned / Given

Subnet ID = Anything

HostID field = All Zero Bits





Topic : Subnet Broadcast Address

→ Used to broadcast a packet to all hosts belongs to a sub-network

→ Special IP address (32 bits)

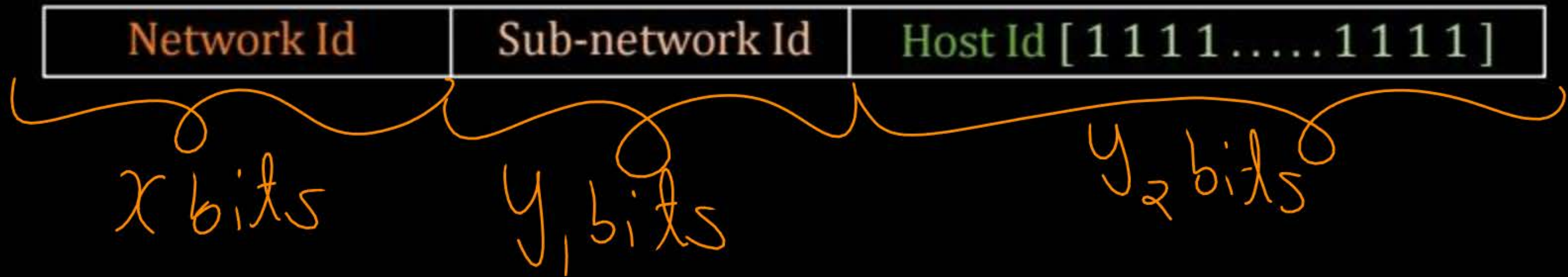


Topic : Subnet Broadcast Address

NetID field = Assigned / Given

Subnet ID = Anything

HostID field = All One Bits



Example 4 :-

#Q. Consider a class-less IPv4 network, the network address of this network is '191 . 191 . 191 . 0 / 24'. The network is divided with 2-bit subnetting. Write subnet address and broadcast address of all the subnets belongs to this network?

→ Network Address = 191.191.191.0/24
Broadcast Add. = 191.191.191.255/24

Special IP

Network Address : 191.191.191.0 / 24

Broadcast Address : 191.191.191.255 / 24

24 bit (prefix)

Net ID

8 bit host ID



For network

prefix = Net ID

Network Address : 191.191.191.0 / 24

Broadcast Address : 191.191.191.255 / 24

Prefix

Before Subnetting :-

First Host IP : 191.191.191.1 / 24

Last Host IP : 191.191.191.254 / 24

$(2^8 - 2)$ IP Addresses.
= 254

Network Address : 191.191.191.0 / 24

Broadcast Address : 191.191.191.255 / 24

Before Subnetting :-

First Host IP : 191.191.191.1 / 24

Last Host IP : 191.191.191.254 / 24

Network Size = $(2^8 - 2) = 254$ IP Address.
for hosts



After 2-bit Subnetting :-

NetID (24) 2 bit SubnetID
 $\Rightarrow$ (old prefix) $\rightarrow$ hostID (6 bits)

Subnet Address

: 191.191.191. _ _ 000000 / 26

First Host IP

: 191.191.191. _ _ 000001 / 26

Last Host IP

: 191.191.191. _ _ 111110 / 26

Subnet Broadcast

: 191.191.191. _ _ 111111 / 26

New prefix
(26 bits)

For subnets

prefix = NetID & SubnetID

First Subnet :-

Subnet Address : 191.191.191. 00 000000 / 26

First Host IP : 191.191.191. 00 000001 / 26

Last Host IP : 191.191.191. 00 111110 / 26

Subnet Broadcast : 191.191.191. 00 111111 / 26

First Subnet :-

→ Prefix

Subnet Address : 191.191.191. 0 / 26

First Host IP : 191.191.191. 1 / 26

Last Host IP : 191.191.191. 62 / 26

} $(2^6 - 2)$ IP Add.
for hosts

Subnet Broadcast : 191.191.191. 63 / 26

Second Subnet :-

Subnet Address : 191.191.191. 0 1 000000 / 26

First Host IP : 191.191.191. 0 1 000001 / 26

Last Host IP : 191.191.191. 0 1 111110 / 26

Subnet Broadcast : 191.191.191. 0 1 111111 / 26

Second Subnet :-

Subnet Address : 191.191.191.64 / 26

First Host IP : 191.191.191.65 / 26

Last Host IP : 191.191.191.126 / 26

} 62 IP Address
for host

Subnet Broadcast : 191.191.191.127 / 26

Third Subnet :-

Subnet Address : 191.191.191. 1 0 000000 / 26

First Host IP : 191.191.191. 1 0 000001 / 26

Last Host IP : 191.191.191. 1 0 111110 / 26

Subnet Broadcast : 191.191.191. 1 0 111111 / 26

Third Subnet :-

Subnet Address : 191.191.191.128 / 26

First Host IP : 191.191.191.129 / 26

Last Host IP : 191.191.191.190 / 26

} 62 IP Add.
(for hosts)

Subnet Broadcast : 191.191.191.191 / 26

Fourth Subnet :-

Subnet Address : 191.191.191. 000000 / 26

First Host IP : 191.191.191. 000001 / 26

Last Host IP : 191.191.191. 111110 / 26

Subnet Broadcast : 191.191.191. 111111 / 26

Fourth Subnet :-

Subnet Address : 191.191.191.192 / 26

First Host IP : 191.191.191.193 / 26

Last Host IP : 191.191.191.254 / 26

$(2^6 - 2)$ IP Addresses
(for hosts)

Subnet Broadcast : 191.191.191.255 / 26

0, 1, 2, - - - - - 254, 255



254 IP Addresses
for host (before subnet)

2-bit subnetting (After subnetting)

Subnet₀₀ → 0, 1, - - - - - , 62, 63

Subnet₀₁ → 64, 65, - - - - - , 126, 127

Subnet₁₀ → 128, 129, - - - - - , 190, 191

Subnet₁₁ → 192, 193, - - - - - , 254, 255



Topic : Subnet Mask



→ Binary string used to produce “Subnet Address” of an IP address



Topic : Subnet Mask



NetID field = All One Bits

Subnet ID = All One Bits

HostID field = All Zero Bits

→ (Prefix)





Topic : Subnet Mask



IP Address :

Network Id

Subnet Id

Host Id

[bit - wise AND]

Subnet Mask :

1 1 1 1

1 1 1 1

0 0 0 0 0 0

Subnet Address :

Network Id

Subnet Id

0 0 0 0 0 0

[MCQ] [GATE-2019] [2-Mark] IIT-M, H.W.



#Q. Consider three machines M, N and P with IP addresses "100 . 10 . 5 . 2", "100 . 10 . 5 . 5" and "100 . 10 . 5 . 6" respectively. The subnet mask is set to "255 . 255 . 252" for all the three machines. Which one of the following is true?

- ☐ A M, N and P all belong to the same subnet
- ☐ B Only N and P belong to the same subnet
- ☐ C Only M and N belong to the same subnet
- ☐ D M, N, and P belong to three different subnets



2 mins Summary



Topic

Subnetting





THANK - YOU